MANISH KUMAR

manish.chaudhary.mrt@gmail.com 9911696464 Gurgaon, India

Professional Summary

Senior Solution Architect with 17+ years of experience driving enterprise-scale data, AI, and analytics platforms. Proven expertise in defining data strategy, leading BI and analytics transformation, and building modern cloud-native data ecosystems using Databricks, AWS, and event-driven architectures. Specialized in Agentic AI systems, LLM platforms, and ML-driven automation, with hands-on experience designing multi-agent workflows, RAG pipelines, and AI-powered decision systems. Strong track record of leading cross-functional teams, influencing architecture decisions, and delivering scalable, production-grade platforms aligned with business outcomes.

Core Expertise

Data Strategy & Architecture
- Enterprise Data Platforms (Databricks, AWS, Data Warehousing)
- Data Governance, Data Quality, Semantic Layer Design
- Data Modeling (Star/Snowflake, SCD, Enterprise-scale systems)

BI, Analytics & Enablement
- BI & Analytics CoE enablement
- Dashboarding, KPI frameworks, self-service analytics
- Data-driven decision systems

Programming
Python, Java, Unix Shell, SQL, Perl

AI / ML & Agentic Systems
- Agentic AI workflows & multi-agent architectures
- LLM Integration (ChatGPT, Gemini, Copilot, open-source models)
- RAG (Retrieval-Augmented Generation) pipelines
- ML models: regression, classification, clustering, anomaly detection

Cloud, Platform & Engineering
- Databricks, PySpark, AWS (SageMaker, Glue, Kinesis, EKS)
- Kafka-based event-driven systems
- Kubernetes, Docker, CI/CD (GitHub Actions, Jenkins)

Technical Leadership

- Led teams of 10+ engineers across data, ML, and platform engineering
- Influenced architecture decisions and enterprise standards
- Acted as technical authority in ML, AI systems, and data platforms
- Mentored teams on ML best practices, engineering quality, and production readiness

Key Data Engineering & ML Projects

- Architected scalable **data platforms and ETL/ELT pipelines** using Databricks and Teradata.
- Designed enterprise **data warehouse and integration solutions** for analytics and reporting.
- Built **ML-based fraud detection** solutions using clustering and unsupervised learning techniques.
- Developed high-volume data processing solutions with focus on **performance, scalability, data quality, and reliability**.
- Applied **machine learning and advanced analytics** to identify patterns, anomalies, and business risks.

Key AI Projects

Enterprise GenAI & Agentic AI Platform — MCP, RAG & AWS Bedrock
- Architected an enterprise GenAI & Agentic AI platform integrating AWS Bedrock, AgentCore, RAG, MCP, and enterprise data/action systems for intelligent analytics, knowledge retrieval, and workflow automation.

RAG-Based Enterprise Knowledge System
- Designed an enterprise RAG architecture combining LLMs, embedding models, vector search, curated knowledge sources, and retrieval pipelines for grounded AI responses.
- Established a scalable knowledge-retrieval layer that reduces reliance on model-only reasoning and improves the accuracy, traceability, and relevance of enterprise AI responses.

AI/LLM Observability & Cost Optimisation

- Designed AI/LLM observability capabilities to monitor agent activity, model usage, token consumption, runtime performance, and operational trends across the enterprise AI platform.
- Implemented caching, session-aware context management, retrieval optimization, and selective tool invocation to reduce unnecessary token usage, improve response efficiency, and optimize GenAI operating costs.

Professional Experience

Senior Solution Architect II, *Adidas* 10/2021 – Current
Gurgaon, India

Leading architecture and development of large-scale AI-driven data platforms and real-time analytics systems for global e-commerce. Key Contributions:

- Defined and implemented data platform strategy using Databricks and AWS, enabling scalable analytics and ML workloads
- Architected real-time fraud detection platform using Kafka and ML pipelines
- Designed agentic AI systems and LLM-powered platforms for intelligent automation and decision support
- Built RAG-based enterprise knowledge systems integrating logs, metrics, and operational data
- Developed AI-driven observability platform for anomaly detection and root cause analysis
- Enabled AI-assisted development workflows (GitHub Copilot, CI/CD automation), improving engineering productivity
- Led data platform modernization and migration to cloud-native architectures

Impact:

- Improved fraud detection accuracy and reduced manual effort
- Accelerated release cycles through AI-driven automation
- Increased system observability and operational efficiency

Senior Software Engineer/Tech Lead, *Optum (UHG)* 08/2014 – 10/2021
Noida, India

Led multiple enterprise data and analytics initiatives in healthcare domain. Key Contributions:

- Led development of enterprise data warehouse (Teradata) supporting organization-wide reporting
- Designed Kafka-based real-time alerting system with microservices architecture
- Built ML-based chatbot (NLP + predictive models) for automation and improved user experience
- Implemented feature engineering, model training, validation, and tuning for large-scale datasets
- Optimized ETL pipelines, reducing execution cost and improving performance
- Designed reusable frameworks for data processing, security, and automation
- Acted as Technical Review Board member, ensuring production readiness

Earlier Experience, *HCL Tech | Wipro | TCS* 10/2008 – 07/2014

- Delivered data warehousing and ETL solutions across banking, healthcare, and e-commerce
- Built scalable backend systems and recommendation engines
- Developed and optimized large-scale data pipelines

Education

M.Tech | Data Analytics, *BITS Pilani*

B.Tech | Information Technology, *UPTU*

Certifications

- Azure Solution Architect
- DataStage Developer
- Oracle Certified Associate (OCA)
- DB2 Developer

Accomplishments and Awards

Recognition from CIO, *Adidas* 2024
For achieving 10+ million Euro in fraud savings

Super Star, *Adidas* 2023

Make IT Happen, *Optum* 01/12/2019

Bright Idea Award, *Optum* 2019
Developed a tool saving approximately $6M.

Top Trainer, *HCL*
Recognized as one of the Top 3 trainers globally within HCL.